

Google I/O '25 Keynote

Overview

Google I/O '25 Keynote is a 1-hour 55-minute Google livestream from May 20, 2025 spanning the company's AI announcements across models, infrastructure, agents, Search, the Gemini app, generative media, and Android XR. Sundar Pichai opens with momentum stats, Demis Hassabis and Tulsi cover Gemini 2.5 and developer features, Liz Reid presents AI Mode in Search, Josh Woodward demonstrates the Gemini app, Jason and Josh introduce generative-media releases including the Flow filmmaking tool, and Sriram and Nishtha demo Android XR including Samsung's Project Muhaan headset and prototype glasses with Gentle Monster and Warby Parker.

Topics covered

- Google has announced over a dozen models and research breakthroughs and released over 20 major AI products and features since the last I/O.
- Elo scores are up more than 300 points since the first generation of Gemini Pro, and Gemini 2.5 Pro sweeps the Elo Arena leaderboard.
- The updated 2.5 Pro is number one on Web Dev Arena, surpassing the previous version by 142 Elo points; on Cursor, Gemini is the fastest growing model of the year.
- Gemini completed Pokémon Blue, earning all eight badges, traveling Victory Road, and defeating the Elite Four and the Champion.
- The 7th generation TPU Ironwood is the first designed for thinking and inference at scale, delivering 10x performance and 42.5 exaflops per pod, coming to Google Cloud customers later this year.
- Google now processes 480 trillion monthly tokens, up from 9.7 trillion a year ago — a roughly 50x increase in a year.
- Over 7 million developers have built with the Gemini API across Google AI Studio and Vertex AI; Gemini usage on Vertex AI is up more than 40 times.
- The Gemini app has over 400 million monthly active users, with 2.5 Pro usage up 45%; AI overviews have more than 1.5 billion users every month.
- Google Beam is a new AI-first video communications platform that turns 2D streams into 3D using six cameras and near-perfect head tracking at 60 fps; first devices with HP later this year.
- Real-time speech translation is launching in Google Meet — English and Spanish for subscribers now, more languages in the coming weeks, enterprises later this year.
- Gemini Live with Project Astra camera and screen sharing rolls out free to everyone on Android and iOS today.
- Project Mariner adds multi-tasking up to 10 simultaneous tasks and a teach-and-repeat feature; computer use is coming to developers via the Gemini API, with Automation Anywhere and UiPath testing, and broader availability this summer.
- Google's open agent-to-agent protocol launched at Cloud Next with the support of over 60 technology partners, and the Gemini SDK is now compatible with Anthropic's Model Context Protocol.
- Agent Mode in the Gemini app uses Project Mariner for apartment hunting on sites like Zillow; the experimental version is coming soon to subscribers.
- Personalized smart replies in Gmail use Gemini personal context drawing from Drive, past emails, and Docs, available in Gmail this summer for subscribers.
- Gemini 2.5 Pro tops Web Dev Arena, incorporates Learn LM for learning, and is number one across all LM Arena leaderboards.
- The updated 2.5 Flash is second only to 2.5 Pro on LM Arena and will be generally available in early June, with 2.5 Pro to follow.
- New 2.5 features include multi-speaker text-to-speech in over 24 languages, native audio dialogue in the Live API, strengthened protections against indirect prompt injections, thought summaries, 22% efficiency gains in Flash, and thinking budgets coming to 2.5 Pro.
- Jules, an asynchronous coding agent that integrates with GitHub, is now in public beta at jules.google.

- Gemini Diffusion is an experimental text diffusion model that generates five times faster than 2.0 Flashlight while matching its coding performance.
- Deep Think is a new 2.5 Pro mode using parallel thinking techniques, with strong results on USAMO 2025, Live Code Bench, and MMMU, going to trusted testers first.
- Google is extending Gemini into a world model, building on Genie 2, and has built Gemini Robotics to teach robots to grasp, follow instructions, and adjust to novel tasks.
- Project Astra demos show the universal AI assistant fixing a bike using voice, web search, video, email, and phone calls.
- Recent Google DeepMind science work includes AlphaProof, Co-scientist, AlphaEvolve, AMY, AlphaFold 3, and Isomorphic Labs; AlphaFold is used by more than 2.5 million researchers worldwide.
- Google partnered with Aira to use Astra technology to help the blind and low-vision community navigate the world.
- AI overviews are in more than 200 countries and territories, driving over 10% growth in their query types in US and India, with Google Lens up 65% year over year and over 100 million visual searches this year.
- AI Mode is a reimagined search experience using Gemini 2.5; it rolls out as a new tab to everyone in the US starting today.
- AI Mode uses a query fan-out technique across the knowledge graph, shopping graph, and Maps community of over 500 million contributors; personal context arrives this summer, deep search soon, and Project Mariner agentic actions are coming for tickets, restaurant reservations, and local appointments.
- Search Live brings Project Astra's live camera into AI Mode, used in a kid-friendly elephant-toothpaste demo.
- Google Lens has over 1.5 billion users every month; Search shopping has over 50 billion product listings, a custom-trained virtual try-on model, and an agentic checkout that tracks price and buys with Google Pay.
- A custom Gemini 2.5 is coming to AI Overviews and AI Mode later this week.
- Gemini Live works in over 45 languages and 150+ countries with conversations five times longer than text; camera and screen sharing roll out free today, and connections to Calendar, Maps, Keep, and Tasks are coming.
- Deep Research now lets users upload their own files, soon across Drive and Gmail; Canvas turns reports into webpages, infographics, quizzes, or custom podcasts in 45 languages.
- Gemini in Chrome is rolling out this week to Gemini subscribers in the US.
- Imagen 4 is Google's new image model with richer detail and better text and typography; the fast variant is 10x faster than the previous model.
- Veo 3 is Google's new state-of-the-art video model with native audio generation — dialogue, sound effects, and background sounds — available today.
- Lyria 2 generates high-fidelity music with vocals, solos, and choirs and is available today for enterprises, YouTube creators, and musicians.
- Over 10 billion pieces of content have been watermarked with SynthID, and a new SynthID Detector for images, audio, text, and video rolls out to early testers today.
- Google partnered with Darren Aronofsky's Primordial Soup to shape Veo as a filmmaking tool, with three short films planned; Eliza McNitt's Ancestor combines live-action with Veo-generated video.
- Flow is a new AI filmmaking tool combining Veo, Imagen, and Gemini, with character and scene consistency and precise camera controls, launching today.
- Google is upgrading subscriptions to Google AI Pro (global) and Google AI Ultra (US today, global soon); Ultra includes Deep Think, Flow with Veo 3, YouTube Premium, and storage.
- Android XR is the first Android platform built in the Gemini era, supporting headsets to glasses, built with Samsung and optimized for Snapdragon with Qualcomm.
- Samsung's Project Muhan headset is the first Android XR device and will be available later this year.
- Android XR glasses are designed for all-day wear with camera, microphones, speakers, and an optional in-lens display; live chat translation was demoed in Hindi and Farsi.
- Glasses prototypes are in trusted-tester hands; developers can build for glasses later this year; Gentle Monster and Warby Parker are the first eyewear partners.
- Fire Sat is a satellite constellation using multi-spectral imagery and AI to detect fires as small as 270 square feet, with imagery updated every 20 minutes when fully operational instead of every 12 hours.
- During Hurricane Helene, Wing partnered with Walmart and the Red Cross to deliver

food and medicine to a YMCA shelter in North Carolina.

People, organizations, products

- People: Sundar, Demis, Tulsi, Liz, Rajan, Vidhya, Bidya, Josh, Jason, Sriram, Nishtha, Darren Aronofsky, Eliza McNitt, Shankar Mahadevan.
- Organizations: Google, Google DeepMind, Anthropic, HP, Samsung, Qualcomm, Automation Anywhere, UiPath, Cursor, Zillow, Aira, Isomorphic Labs, Primordial Soup, Gentle Monster, Warby Parker, Walmart, Red Cross.
- Products: Gemini, Gemini 2.5 Pro, Gemini 2.5 Flash, Deep Think, Gemini Diffusion, TPU Ironwood, Google Beam, Project Astra, Project Mariner, Agent Mode, Gemini API, Vertex AI, Google AI Studio, Jules, Learn LM, Gemini Live, Gemini Robotics, AlphaFold 3, AlphaProof, Co-scientist, AlphaEvolve, Genie 2, AI Overviews, AI Mode, Google Lens, Search Live, Deep Search, Google Pay, Imagen 4, Véo 3, Lyria 2, SynthID, SynthID Detector, Flow, Google AI Pro, Google AI Ultra, YouTube Premium, Android XR, Project Muhaan, Snapdragon, Fire Sat, Wing, Waymo, Gemini in Chrome, Canvas, Deep Research, Model Context Protocol.

Numbers and data points

- Over a dozen models and research breakthroughs since last I/O.
- Over 20 major AI products and features since last I/O.
- More than 300 Elo points improvement since the first Gemini Pro.
- 142 Elo points by which the updated 2.5 Pro surpasses the previous on Web Dev Arena.
- 10x TPU Ironwood performance over the previous generation.
- 42.5 exaflops of compute per TPU Ironwood pod.
- 9.7 trillion monthly tokens a year ago vs 480 trillion now — about 50x in a year.
- Over 7 million developers on the Gemini API; over 5x growth since last I/O; over 40x growth on Vertex AI.
- Over 400 million monthly active users of the Gemini app; 45% usage increase for 2.5 Pro.
- 1.5 billion monthly users of AI overviews in more than 200 countries and territories.
- Over 10% growth in queries that show AI overviews in markets like US and India.
- 65% year-over-year growth in Google Lens; more than 100 million visual searches this year.
- Google Beam: six cameras, 60 frames per second head tracking.
- Project Mariner: up to 10 simultaneous tasks; over 60 partners at the agent-to-agent protocol launch.
- Gemini text-to-speech: 24+ languages.
- 22% efficiency gains in updated 2.5 Flash.
- 37 seconds: thinking time for the 2.5 Pro 3D sketch-to-code demo.
- Gemini Diffusion: five times faster than 2.0 Flashlight.
- 2.5 million researchers worldwide using AlphaFold.
- Over 1.5 billion monthly users of Google Lens.
- 500 million Maps community contributors.
- Over 50 billion product listings in the shopping graph.
- Gemini Live: 45 languages, more than 150 countries, conversations five times longer than text.
- Imagen 4 fast variant: 10x faster than the previous model.
- 10 billion pieces of content watermarked with SynthID.
- Fire Sat: detects fires as small as 270 square feet; refreshes every 20 minutes vs every 12 hours today.
- Over 10 years building glasses.

Notable quotes

- Sundar: “Every day is Gemini season here at Google.”
- Sundar: “in our Gemini era, we’re just as likely to ship our most intelligent model on a random Tuesday in March or a really cool breakthrough like Alpha Evolve just a week before.”

- Sundar: “Now, we are processing 480 trillion monthly tokens.”
- Sundar: “The Gemini app now has over 400 million monthly active users.”
- Sundar: “Google Search is bringing generative AI to more people than any other product in the world.”
- Sundar: “Introducing Google Beam, a new AI first video communications platform.”
- Sundar: “we think of agents as systems that combine the intelligence of advanced AI models with access to tools.”
- Demis: “We’re living through a remarkable moment in history, where AI is making possible an amazing new future.”
- Demis: “Gemini 2.5 Pro is our most intelligent model ever, and the best foundation model in the world.”
- Tulsi: “I’m thrilled to announce that we’re releasing an updated version of 2.5 Flash.”
- Demis: “Today, we’re making 2.5 Pro even better by introducing a new mode we’re calling Deep Think.”
- Demis: “we’re working hard to extend it to become what we call a world model.”
- Demis: “To transform it into a universal AI assistant. An AI that’s personal, proactive, and powerful.”
- Sundar: “AI mode is coming to everyone in the US starting today.”
- Liz: “This is the future of Google Search, a search that goes beyond information to intelligence.”
- Liz: “AI mode is Search transformed with Gemini 2.5 at its core.”
- Liz: “We couldn’t be more excited about this chapter of Google Search where you can truly ask anything.”
- Josh: “It’s called Imagine 4 and it’s a big leap forward.”
- Josh: “V O 3 comes with native audio generation.”
- Jason: “generative media is expanding the boundaries of creativity.”
- Josh: “We’re calling it Flow, and it’s launching today.”
- Sriram: “It’s the first Android platform built in the Gemini era”
- Sriram: “Gentle Monster and Warby Parker will be the first eyewear partners to build glasses with Android XR.”
- Sundar: “Together with an amazing group of partners, we are building something called Fire Sat.”
- Sundar: “It can detect fires as small as 270 square feet, about the size of a one-car garage.”